13 $(6^{-2})^w = 6^8$

(a) Find the value of w.

$w =$ .....

(1)

$\dfrac{(a^r)^2}{(a^t)^3}$ can be written in the form a^u

(b) Find an expression for u in terms of r and t.

.....

(2)

Given that $n^{\frac{2}{3}} = 8$ and $n > 0$

(c) work out the value of n.
Give your answer in the form $a\sqrt{b}$ where a and b are integers.

.....

(2)

(Total for Question 13 is 5 marks)